

SAI RAM REDDY SOLLETI

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SUMMARY

Applied Artificial Intelligence master's student at Purdue University (4.0 GPA), building end-to-end machine learning systems, from conversational voice agents to multilingual speech and vision models. Skilled in Python, PyTorch, TensorFlow, and Scikit-learn, with hands-on experience deploying containerized ML services to AWS. Seeking AI/ML engineering roles where research translates into deployed, measurable systems.

EDUCATION

Purdue University — M.S., Applied Artificial Intelligence *Jan 2025 – Dec 2026 (Expected)*
GPA 4.0 / 4.0

TECHNICAL SKILLS

Programming: Python, SQL, Java, C, JavaScript, PHP
ML & Deep Learning: PyTorch, TensorFlow, Scikit-learn, XGBoost, Hugging Face Transformers, OpenCV, LLM fine-tuning, RAG
Voice & Agents: Twilio, Deepgram, ElevenLabs, OpenAI Whisper, n8n
Data & Analysis: Pandas, NumPy, Matplotlib, Seaborn, SHAP, Jupyter, Anaconda
Cloud & MLOps: AWS (EC2, S3), Azure, Docker, Docker Compose, Cloudflare Tunnels, Git, GitHub Actions
Databases: MySQL, PostgreSQL, MongoDB, Elasticsearch
Web & APIs: FastAPI, Flask, React.js, Node.js, Express, REST

PROJECTS

Hotel Voice Agent — Conversational Booking System *2026*
Built a phone line that takes hotel bookings: Twilio for telephony, Deepgram Nova-2 for speech, an LLM tool-calling layer over FastAPI and PostgreSQL, ElevenLabs for the reply, wired in n8n. Completed 95.6% of tasks across 90 staged calls at 4.8 s median turn latency.

Quantum-Secured Voting System — Quantum Machine Learning *2026*
Implemented a voting platform with genuinely quantum randomness. Ballot receipts draw 96 bits from 16-qubit Hadamard circuits in Qiskit, alongside a BB84 key-distribution protocol; voter identity is decoupled by SHA-256 commitment, events by a hash-chained audit log.

Road Sign Classification — Transfer Learning with ResNet-50 *2026*
Fine-tuned ResNet-50 on GTSRB in two phases, training the classification head against a frozen backbone before unfreezing the whole network at a lower learning rate. Final numbers: 98.46% test accuracy and 0.9845 weighted F1 across 43 classes on 12,630 held-out images.

Multilingual Transliteration Chat — with Whisper Voice Input *2025*
Shipped a chat app that lets users talk to an LLM in ten non-Latin scripts, transliterating in the browser so raw text never leaves the client; faster-whisper voice input came later. Scored 93.4% exact-match accuracy on a 1,000-prompt corpus at 1.2 s median round trip.

Transformer Time-Series Forecasting *2025*
Trained a small decoder-only transformer (128-dim embeddings, 4 heads, 3 layers) on 1,748 Bitcoin sequences drawn from 87,000 records via 40-step sliding windows. R^2 0.9213, MAE 0.1087 on held-out data.

Lang2Code — Natural Language to Python Code Generation *2024*
Created a system that turns plain-English requests into Python, using a custom spaCy NER model with GloVe embeddings trained on 525 annotated queries across 20 intents and 10 entity types, mapping onto a library of 100 snippets behind a REST API.

CERTIFICATIONS

AWS Certified AI Practitioner , Amazon Web Services · 2026	Python 3 Programming , Univ. of Michigan / Coursera · 2023
CCNA (Cisco Certified Network Associate) , Cisco · 2024	Cybersecurity Essentials , Cisco · 2023
Java Full Stack , Wipro · 2023	Introduction to IoT , Cisco · 2022

EXPERIENCE

Research Assistant — Purdue University *Aug 2025 – May 2026*
Worked with Dr. Ashok Vardhan Raja on conversational AI and applied machine learning, taking the Hotel Voice Agent from implementation through evaluation design, measurement, and technical write-up.

Genpact — Process Associate *Jul 2024 – Sep 2024*
Resolved customer queries with cross-functional teams, cutting average response time by 15%, and supported rollout of new process policies and internal systems.